

PRACTICAL_NO_8

AIM: Implementation of DSR Routing Protocol

Understanding DSR in NS2

- Reactive Protocol: DSR finds routes only when a source node needs to send data.
 - Source Routing: The complete path from source to destination is stored in the packet header.
 - Built-in Support: NS2 includes native C++ and OTcl implementations for DSR in its wireless modules
- Steps to Configure DSR in a TCL Script

```
... (File "dsr_sim.tcl" line 1)
mvl@IT-09:~$ nano dsr_sim.tcl
mvl@IT-09:~$ cat dsr_sim.tcl
set ns [new Simulator]
$ns node-config -adhocRouting DSR \
  -llType LL \
  -macType Mac/802_11 \
  -ifqType Queue/DropTail/PriQueue \
  -ifqLen 50 \
  -antType Antenna/OmniAntenna \
  -propType Propagation/TwoRayGround \
  -phyType Phy/WirelessPhy \
  -channelType Channel/WirelessChannel \
  -topoInstance $topo \
  -agentTrace ON \
  -routerTrace ON \
  -macTrace OFF
mvl@IT-09:~$ nano dsr_sim.tcl
mvl@IT-09:~$ ls
aodv_out.nam  aodv_sim.tcl  dsr.tr        dsr_sim.tcl   nam_1.15-6.dsc  nam_error.txt
aodv_out.tr   dsr.nam       dsr_practical  nam_1.15-6.debian.tar.xz  nam_1.15.orig.tar.gz  'nano aodv_sim.tcl'
mvl@IT-09:~$ nam dsr_out.nam
nam:
mvl@IT-09:~$ nam dsr.nam
nam:
mvl@IT-09:~$ wc -l dsr.nam
0 dsr.nam
mvl@IT-09:~$ head -20 dsr.nam
mvl@IT-09:~$ nano dsr_sim.tcl
mvl@IT-09:~$ rm -f dsr.nam dsr.tr
mvl@IT-09:~$ ns dsr_sim.tcl
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
can't read "phyType.": no such variable
while executing
"$node add-interface $chan $propInstance_ $llType_ $macType_ $ifqType_ $ifqLen_ $phyType_ $antType_ $topoInstance_ $inerrProc_ $outerrProc_ $fecProc_"
(procedure "c_03" line 83)
(Simulator create-wireless-node line 83)
invoked from within
"o3 create-wireless-node"
("eval" body line 1)
invoked from within

GNU nano 7.2 dsr_sim.tcl
set ns [new Simulator]

set tracefd [open dsr.tr w]
set namfd [open dsr.nam w]

set topo [new Topography]
$topo load_flatgrid 500 500

set god_ [create-god 3]

$ns trace-all $tracefd
$ns namtrace-all-wireless $namfd 500 500

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
  -adhocRouting DSR \
  -llType LL \
  -macType Mac/802_11 \
  -ifqType Queue/DropTail/PriQueue \
  -ifqLen 50 \
  -antType Antenna/OmniAntenna \
  -propType Propagation/TwoRayGround \
  -phyType Phy/WirelessPhy \
  -channel $chan_1_ \
  -topoInstance $topo \
  -agentTrace ON \
  -routerTrace ON \
  -macTrace OFF

set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0
```

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SUBJECT: WIRELESS SENSOR NETWORKS

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```
GNU nano 7.2 dsr_sim.tcl
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0

$ns initial_node_pos $n0 50
$ns initial_node_pos $n1 150
$ns initial_node_pos $n2 250

set udp [new Agent/UDP]
$ns attach-agent $n0 $udp

set null [new Agent/Null]
$ns attach-agent $n2 $null

$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr set packetSize 512
$cbr set interval 0.1
$cbr attach-agent $udp

$ns at 1.0 "$cbr start"
$ns at 8.0 "$cbr stop"

proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    exit 0
}

$ns at 10.0 "finish"
```

```
invoked from within
"God instance"
(procedure "_o13" line 113)
(Object next line 113)
invoked from within
"_o13 next _o11 _o12 ll Mac/802_11 Queue/DropTail/PriQueue 50 Phy/WirelessPhy Antenna/OmniAntenna _o10 {} {} {}"
("eval" body line 1)
invoked from within
"eval $self next $args"
(procedure "_o13" line 2)
(SRNodeNew add-interface line 2)
invoked from within
"$node add-interface $chan $propInstance_ $llType_ $macType_ $ifqType_ $ifqlen_ $phyType_ $antType_ $stopInstance_ $inerrProc_ $outerrProc_ $fecProc_"
(procedure "_o3" line 83)
(Simulator create-wireless-node line 83)
invoked from within
"_o3 create-wireless-node"
("eval" body line 1)
invoked from within
"eval $self create-wireless-node $args"
(procedure "_o3" line 23)
(Simulator node line 23)
invoked from within
"$ns node"
invoked from within
"set n0 [$ns node]"
(file "dsr_sim.tcl" line 29)
mvl@IT-09:~$ nano dsr_sim.tcl
mvl@IT-09:~$ rm -f dsr.nam dsr.tr
ns dsr_sim.tcl
num_nodes is set 3
INITIALIZE THE LIST xlistHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
mvl@IT-09:~$ wc -l dsr.nam
510 dsr.nam
mvl@IT-09:~$ nam dsr.nam
nam:
mvl@IT-09:~$ which nam
/usr/bin/nam
mvl@IT-09:~$ nam -v
```